

Foundation (Jalapeno)

Q1. Simplify $2^3 \cdot 2^4$

- (A) 2^{3+4}
- (B) $2^{3 \cdot 4}$
- (C) 2^{3-4}
- (D) 2^{4-3}
- (E) $2^{3/4}$

Q2. Simplify $(3^2)^3$

- (A) 3^{2+3}
- (B) $3^{2 \cdot 3}$
- (C) 3^{2-3}
- (D) 3^{3-2}
- (E) $3^{2/3}$

Q3. Simplify $\frac{4^3}{4^2}$

- (A) 4^{3+2}
- (B) $4^{3 \cdot 2}$
- (C) 4^{3-2}
- (D) 4^{2-3}
- (E) $4^{3/2}$

Q4. Simplify $\frac{5^2}{5^4}$

- (A) 5^{2+4}
- (B) $5^{2 \cdot 4}$
- (C) 5^{2-4}
- (D) 5^{4-2}
- (E) $5^{2/4}$

Q5. Simplify $(2^3)^2$

- (A) 2^{3+2}
- (B) $2^{3 \cdot 2}$
- (C) 2^{3-2}
- (D) 2^{2-3}
- (E) $2^{3/2}$

Q6. Simplify $3^4 \cdot 3^2$

- (A) 3^{4+2}
- (B) $3^{4 \cdot 2}$
- (C) 3^{4-2}
- (D) 3^{2-4}
- (E) $3^{4/2}$

Q7. Simplify $\frac{2^5}{2^3}$

- (A) 2^{5+3}
- (B) $2^{5 \cdot 3}$
- (C) 2^{5-3}
- (D) 2^{3-5}
- (E) $2^{5/3}$

Q8. Simplify $(4^2)^3$

- (A) 4^{2+3}
- (B) $4^{2 \cdot 3}$
- (C) 4^{2-3}
- (D) 4^{3-2}
- (E) $4^{2/3}$

Open Ended Questions (FRQ)

Q9. Simplify the expression $\frac{(2^3)^2}{2^4}$ and show your work.

Q10. Simplify the expression $3^2 \cdot 3^4$ and show your work.

Chef's Tips

- Read each question carefully.
- Choose the correct answer from the options provided.
- Show work for free-response questions.